Daniel Nguyen

Australian Citizen | Website | daniel.ngu05@gmail.com | Linkedin | Github

Experience

Digital Engineer October 2025 – May 2026

Linquist

- Integrated third-party AI APIs such as OpenAI and ElevenLabs to automate content and voice generation whilst supporting PDF rendering workflows and reducing manual content creation time by 80%.
- Implemented credit management middleware with validation layers and protected REST API endpoints to enforce subscription based rate limits, securing 100% of monetised actions at the server layer.
- Developed over 20 type-safe frontend features using Next.js and TypeScript, including exercise filtering, pagination, and role-based UI rendering on a GCP-hosted platform with Stripe payment integration.

AI Research Assistant (Security Regressions in LLM-Refactored Code) March 2026 – Present

Monash University

- Built an automated pipeline prompting multiple LLMs (GPT-4, Claude) across 100+ Java/C++ functions with varied refactoring instructions, using CodeQL and Semgrep to detect newly introduced CWEs in outputs.
- Validated functional correctness of all refactored code via unit test suites, identifying functionality drift across 20% of cases and categorising which prompt types posed the greatest security risk.
- Published empirical findings on the real-world safety of AI coding tools such as GitHub Copilot and Cursor, revealing that LLM refactoring introduced silent security vulnerabilities in 12% of previously secure codebases

Founder August 2026 – Present

Veltris Labs

- Architected an automated B2B SaaS verification platform for government energy-rebate certification, integrating with ServiceM8 and Tradify to eliminate manual cross-checking across 70+ fields per job performed by accredited auditors.
- Designed a custom ServiceM8 plugin to preserve photo geotagging metadata through the upload pipeline, resolving a systemic data-integrity issue causing silent location-data loss on transfer to downstream systems such as Salesforce.
- Built a scalable, webhook-driven automation pipeline (n8n) orchestrating multistep end-to-end verification workflows, designed to seamlessly support high-volume concurrent submissions across a pipeline of 200+ target installer businesses.

Education

Monash University Melbourne, Australia

Bachelor of Computer Science Advanced, Grade: High Distinction *Feb 2024 – Nov 2026*

University of Texas at Austin Austin, Texas

Bachelor of Science in Computer Science — Semester Exchange Program *Aug 2025 – Dec 2025*

Projects

Finova Investment Platform | Link

- Designed a scalable financial analytics platform integrating the Alpha Vantage API to process market data (1,000+ data points per request), leveraging caching to reduce redundant API calls and improve latency by 40%.
- Developed an AI financial assistant by deploying a custom Ollama 3.2 model to generate data insights for complex financial queries and embedding financial context to improve response accuracy.
- Built an interactive portfolio optimisation tool leveraging linear programming and predictive analytics, enabling simulation of multi-asset allocations and optimisation of expected returns under configurable risk constraints.

BevoGuard | Link

- Developed a full-stack web application using React and Flask to help new Austin residents monitor neighborhood safety, featuring dynamic map-based visualisation and location-based search functionality.
- Integrated real-time public safety APIs and automated data ingestion pipelines to process 10,000+ incident reports and city alerts, providing up-to-date safety insights.
- Optimised backend performance using Flask-based routing and indexed MySQL queries for efficient data retrieval, deploying the application on AWS for scalable cloud hosting.

Technical Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML/CSS, C++, C#, Haskell, Go, Swift, R
Frameworks & Libraries: React, Next.js, Node.js, Flask, Unity, Selenium, Pygame, NumPy, Pandas, LangChain
Cloud, DevOps: AWS (Lambda, DynamoDB, S3, Cognito), GCP, Docker, Kubernetes, CI/CD, RESTful APIs, Terraform
AI/ML & Security: OpenAI API, Ollama, LLM Integration, Prompt Engineering, CodeQL, Semgrep, OWASP
Developer Tools & Practices: Git, GitHub, GitLab, VS Code, Postman, Agile, SDLC, Unit Testing, PostgreSQL, MySQL